JOHANSSON ENGINEERING

Equipment Lifetime Journal — Gothenburg Workshop

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Client	Example Workshop AB
Site	Gothenburg Workshop

1. Purpose

This journal tracks the **remaining lifetime** of pressurised vessels at Gothenburg Workshop as required by AFS 2017:3, 4 Kap. §18. It records wall thickness measurements, calculates degradation rates, and estimates remaining service life.

> This is an **example document** demonstrating realistic corrosion tracking for the Gothenburg Workshop vessel set. The measurement data is illustrative.

2. Vessel Lifetime Data

2.1 Design Parameters

VESSEL ID	DESCRIPTION	MATERIAL	ORIG. WALL (MM)	MIN. REQ. (MM)	CORR. ALLOW. (MM)
PV-001	Main air receiver	P265GH	8.0	5.2	2.8
PV-003	Steam boiler	16Mo3	12.0	8.5	3.5
PV-004	Refrigeration receiver	P355N	10.0	7.0	3.0
PV-005	Hydraulic accumulator	P265GH	6.0	4.2	1.8

> PV-002 and PV-007 (air receivers, Class B exempt) and PV-006 (below threshold) are not tracked in this journal per §18 scope (Class A and B with limited lifetime only).

2.2 Active Degradation Mechanisms

VESSEL ID	GEN. CORROSION	PITTING	CUI RISK	FATIGUE	CREEP
PV-001	Yes	No	No	No	No
PV-003	Yes	No	Yes	No	Yes
PV-004	Yes	Yes	No	No	No
PV-005	No	No	No	Yes	No

Notes: - **PV-001:** General internal corrosion from moisture in compressed air. External corrosion minimal (indoor, dry). - **PV-003:** Fire-side corrosion + CUI risk on insulated sections. Creep tracking required (operating >110°C). - **PV-004:** Ammonia stress corrosion cracking risk. Pitting possible at weld zones. - **PV-005:** Hydraulic cycling fatigue (not corrosion-limited). Approx. 50 cycles/day.

3. Wall Thickness Measurements

PV-001 — Main Air Receiver (Class A, Compressed Air)

Measurement history:

DATE	LOCATION	MEASURED (MM)	RATE (MM/YR)	REMAINING (YR)	INSPECTOR	METHOD
2015-06-01	Bottom shell	8.0	—	—	DEKRA	UTM (baseline)
2019-06-15	Bottom shell	7.6	0.10	24.0	DEKRA	UTM
2019-06-15	Top shell	7.8	0.05	52.0	DEKRA	UTM
2019-06-15	Drain nozzle	7.4	0.15	14.7	DEKRA	UTM
2024-06-15	Bottom shell	7.1	0.10	19.0	DEKRA	UTM
2024-06-15	Top shell	7.6	0.04	60.0	DEKRA	UTM
2024-06-15	Drain nozzle	6.7	0.14	10.7	DEKRA	UTM

Limiting location: Drain nozzle (highest corrosion rate) **Current corrosion rate:** 0.14 mm/year **Remaining life:** 10.7 years (from 2024 measurement) **Condition trend:** Stable **Action required:** Monitor — drain nozzle rate is elevated but within acceptable range. Ensure condensate drain is operated regularly.

PV-003 — Steam Boiler (Class A, Steam)

Measurement history:

DATE	LOCATION	MEASURED (MM)	RATE (MM/YR)	REMAINING (YR)	INSPECTOR	METHOD
2010-09-01	Fire-side shell	12.0	—	—	Kiwa	UTM (baseline)
2014-09-01	Fire-side shell	11.2	0.20	13.5	Kiwa	UTM
2014-09-01	Water-side shell	11.6	0.10	31.0	Kiwa	UTM
2014-09-01	Tube plate	11.0	0.25	10.0	Kiwa	UTM
2018-09-01	Fire-side shell	10.4	0.20	9.5	Kiwa	UTM
2018-09-01	Water-side shell	11.2	0.10	27.0	Kiwa	UTM
2018-09-01	Tube plate	10.0	0.25	6.0	Kiwa	UTM
2023-09-01	Fire-side shell	9.4	0.20	4.5	Kiwa	UTM
2023-09-01	Water-side shell	10.7	0.10	22.0	Kiwa	UTM
2023-09-01	Tube plate	8.8	0.24	1.3	Kiwa	UTM

Limiting location: Tube plate (highest corrosion rate + lowest remaining life) **Current corrosion rate:** 0.24 mm/year (tube plate) **Remaining life: 1.3 years** (from 2023 measurement) — **CRITICAL Condition trend:** Declining — tube plate degrading faster than predicted **Action required: Engineering assessment required.** Tube plate approaching minimum thickness. Must not be pressurised beyond Q1 2025 without documented lifetime extension analysis or repair. Escalate to operator immediately.

PV-004 — Refrigeration Receiver (Class A, Ammonia)

Measurement history:

DATE	LOCATION	MEASURED (MM)	RATE (MM/YR)	REMAINING (YR)	INSPECTOR	METHOD
2019-03-01	Shell body	10.0	—	—	RISE	UTM (baseline)
2022-03-10	Shell body	9.9	0.03	96.7	RISE	UTM
2022-03-10	Weld seam A	9.8	0.07	40.0	RISE	UTM
2025-03-10	Shell body	9.8	0.03	93.3	RISE	UTM
2025-03-10	Weld seam A	9.7	0.05	54.0	RISE	UTM

Pitting inspection (visual + MPI at weld zones):

DATE	LOCATION	FINDINGS	MAX PIT DEPTH (MM)
2022-03-10	Weld seam A	No pitting found	—
2025-03-10	Weld seam A	Minor surface indications, not measured as pitting	—

Limiting location: Weld seam A **Current corrosion rate:** 0.05 mm/year (very low — ammonia systems typically clean) **Remaining life:** 54 years **Condition trend:** Stable **Action required:** None — continue standard program. Monitor weld seam indications at next examination.

PV-005 — Hydraulic Accumulator (Class B, Hydraulic Oil)

This vessel is fatigue-limited, not corrosion-limited. Wall thickness is not the primary degradation mechanism.

Fatigue tracking:

PARAMETER	VALUE
Design cycles (manufacturer)	500,000
Estimated usage	~50 cycles/day, 250 days/year = 12,500/year
Service start	2020
Cycles to date (est.)	75,000 (6 years x 12,500)
Remaining cycles	425,000
Estimated remaining life	**34 years**

Condition trend: Stable — well within fatigue design limit **Action required:** None — continue standard program. Verify cycle estimate if operating pattern changes.

4. Creep Tracking — PV-003

Steam boiler operates above 110°C — creep tracking required.

PARAMETER	VALUE
Design temperature	180°C
Creep design hours (manufacturer)	200,000 hours
Operating hours to date	~105,000 hours (2010-2026, ~75% duty)
Remaining hours	95,000
Estimated usage rate	~6,500 hours/year
Estimated remaining creep life	**14.6 years**

> **Note:** Creep life is estimated. Actual remaining life is limited by the **tube plate corrosion** (1.3 years) — see Section 3.

5. Lifetime Summary

VESSEL ID	LIMITING MECHANISM	REMAINING LIFE	STATUS	NEXT MEASUREMENT
PV-001	General corrosion (drain nozzle)	**10.7 years**	Safe	2028-06
PV-003	General corrosion (tube plate)	**1.3 years**	**Critical**	**Immediate**
PV-004	General corrosion (weld seam)	54 years	Safe	2029-03
PV-005	Fatigue cycling	34 years	Safe	2030 (review)

Status Legend

STATUS	REMAINING LIFE	ACTION
Safe	> 2x inspection interval	Continue normal program
Monitor	1x — 2x inspection interval	Increase measurement frequency
Critical	< 1x inspection interval	Engineering assessment before next pressurisation
Expired	≤ 0 years	Must NOT be pressurised without lifetime extension analysis (§18)

6. Lifetime Extension — PV-003 Tube Plate

> **This section demonstrates what §18 second paragraph requires when equipment approaches end of life.**

Vessel	PV-003 — Steam boiler
Component	Tube plate
Original end-of-life	Q1 2025 (based on 2023 measurement)
Extension analysis	[To be performed — fitness-for-service assessment per EN 13445-3 Annex B or API 579-1]
New end-of-life	[Pending analysis]
Approved by	[Name, role]

Required actions before continued pressurisation: 1. Commission fitness-for-service assessment by competent engineer 2. Consider repair options (weld overlay, tube plate replacement) 3. If extended: document the analysis, update this journal, inform inspection body 4. If not extendable: take PV-003 out of service, plan replacement

Revision History

REV	DATE	AUTHOR	DESCRIPTION
A	2026-04-10	N. Johansson	Initial journal — 4 vessels tracked. PV-003 tube plate flagged CRITICAL (1.3 yr remaining).